CHAPTER

7 UX Research Data Analysis

Highlights

- Distill the essence from your UX research by composing elemental work activity notes from raw data notes
- Synthesize UX research data for insight and understanding
- Be efficient about modeling
- UX research data are incomplete
- UX research data present multiple overlapping perspectives
- Data models and the work activity affinity diagram (WAAD) are synthesis choices
- Construct a WAAD to organize the UX research data

7.1 INTRODUCTION

7.1.1 You Are Here

We begin each process chapter with a "You are here" picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 7.1). Within the understand-needs UX design lifecycle activity, this chapter is about analysis of the UX research data you gathered in Chapter 6, furthering your goal of understanding the user work[1] practice and work context for the new product or system you will design.

7.1.2 Goal of UX Research Analysis

The goal of UX research analysis is to distill, clarify, organize, and synthesize raw UX research data (the output of UX research data gathering) into a form that is easily comprehensible within the broader goal of understanding design needs for the product or system to be designed, via the lens of user work practice in the work domain.

If the system or product you are working on is large, unfamiliar, and/or domain complex (see Section 3.2.1.3), you can come away from data gathering

[1]The term *work* broadly signifies anything people/users do while using products or systems (see Section 6.2).

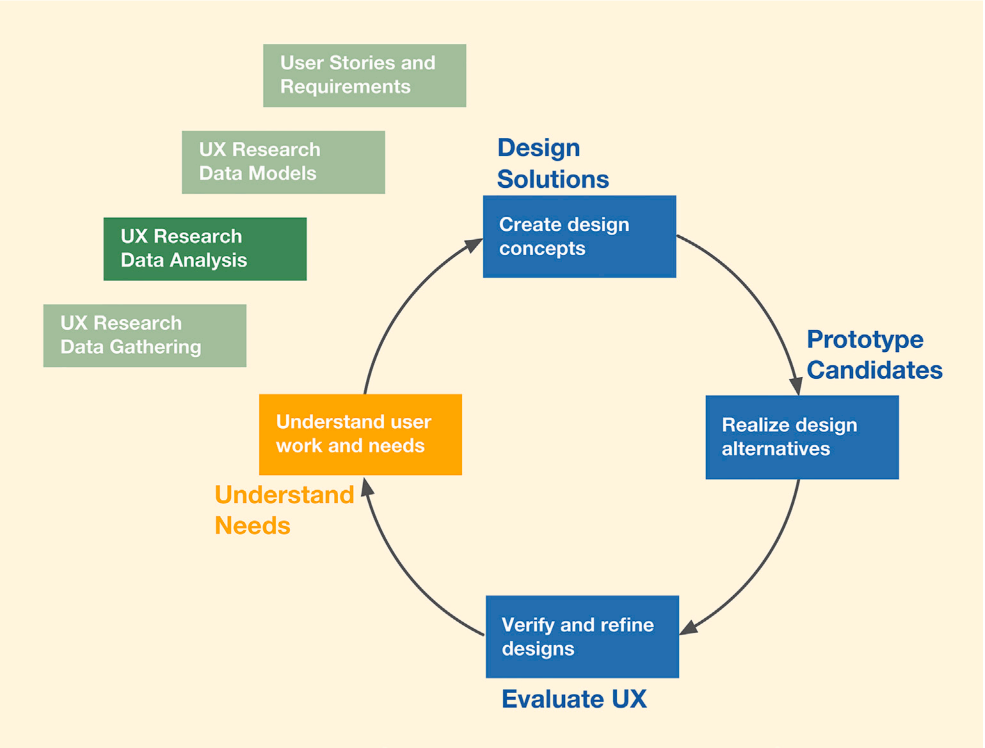

Fig. 7.1

You are here in the UX research data analysis chapter within the understand-needs activity, in the context of the overall Wheel UX lifecycle.

feeling overwhelmed. This is when you can use the power of the UX process to distill and synthesize UX research data.

7.1.3 Preview of the Process

Fig. 7.2 shows a preview of UX research data analysis and its context.

The raw UX research data gathered in Chapter 6 is distilled into work activity notes (Section 7.2), which are synthesized into a work activity affinity diagram (WAAD; Section 7.4) and the various data models (Chapters 8 and 9). These synthesized data representations are used (in Chapter 10) as design informers to create user stories and requirements to drive the UX design.

7.2 DISTILL THE ESSENCE OF YOUR UX RESEARCH DATA BY COMPOSING WORK ACTIVITY NOTES

Start with the set of raw UX research data notes that came from data gathering in Chapter 6 and, for each raw data note, compose one or more work activity notes. As we said in Section 6.7.9, with experience, the UX data

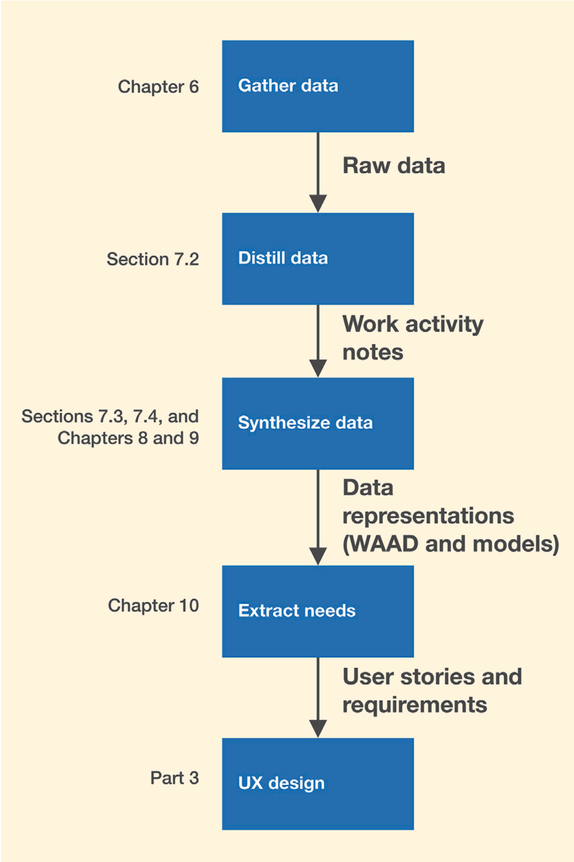

Fig. 7.2
Preview of UX research analysis and its context.

collector understands the nature of work activity notes and can skip over most raw UX research data capture, going directly to final, or near-final, work activity notes.

Each work activity note should make a single point about the work practice that is:

- Elemental
- Clear
- Concise
- Complete
- Modular

Although the process can work if you write work activity notes on scraps of paper or sticky notes, we recommend composing notes in computer-readable form, whether in a word processor, a spreadsheet, or directly into a database system. This facilitates sharing, manipulating, and printing as needed.

7.2.1 Make Each Work Activity Note Elemental

A note is elemental if it refers to or relates to exactly one simple basic concept, idea, fact, or topic. One important point to note is that even though we attempt to capture one concept or topic at a time by keeping the work activity notes purely elemental, the innate overlapping nature of UX research data requires us to recognize that a given topic might still span multiple abstract perspectives (e.g., work roles, user tasks, work context). For example, a work activity note about a user wanting to pay using a credit card is elemental in that it refers to one payment method, but it covers such perspectives as user role (ticket buyer), task (payment), and context (buying tickets).

7.2.1.1 Why elemental?

Years ago, when we began teaching about writing contextual/user/UX research data notes in our classes, the first step was to move directly from raw data notes to the rest of the process. Students were using raw or lightly edited data notes as inputs to the analysis and synthesis activities (e.g., WAADs; Section 7.4), and it was not working. The notes were too verbose and scattered because they were direct quotes from users and covered multiple topics with no order. The team often had to read each note multiple times to understand its meaning. We spent days with the teams outside class trying to get their affinity diagrams and other analysis activities finished and were often not successful. The key to overcoming this barrier to the following analysis and synthesis activities was to scrub the raw UX research data by making the work activity notes elemental, clear, concise, complete, and modular. We cover these themes in the following sections.

Example: Composing Elemental Work Activity Notes for the Ticket Kiosk System, Part 1

Consider this raw data note:

> *It is too difficult to get enough information about events from a ticket seller at the ticket window. I would like to be able to find my own events and not depend on the ticket seller to do the browsing and searching.*

There are at least two elemental work activity notes that can be composed simply by breaking apart the two sentences:

It is too difficult to get enough information about events from a ticket seller at the ticket window.

I would like to be able to find my own events and not depend on the ticket seller to do the browsing and searching.

These two sentences may be condensed to:

I want to be able to browse and search for my own events.

Example: Composing Elemental Work Activity Notes for the Ticket Kiosk System, Part 2

Another example of composing elemental notes for the TKS considers this raw data note:

I like to keep up to date with what is going on regarding entertainment in the community. I especially want to know about current and popular events.

Again, we can make at least two work activity notes simply by breaking apart the two sentences:

I like to keep up to date with community entertainment.

I especially want to know about current and popular events.

7.2.2 Clarify Each Work Activity Note

Clarify each work activity note by:

- Making it easy to read and understand at a glance
- Making a focused point, conveying the substance of the issue in question
- Avoiding vague terms
- Resolving ambiguities

Example: Composing a Clear Work Activity Note for the Ticket Kiosk System

Consider this work activity note again:

I like to keep up to date with community entertainment.

Perhaps the thought behind this work activity note would be clearer and direct if it said:

I would like to see listings of current community entertainment.

7.2.3 Make Each Work Activity Note Concise

Make each work activity note brief and to the point.

- Filter out all noise, fluff, and non-essential verbiage; remove the chaff from the wheat.
- Rephrase, condense, and distill; boil it down to the essence.

Example: Composing Concise Work Activity Notes for the Ticket Kiosk System

Consider this user comment in the raw data:

> *It is difficult to get enough information about events from a ticket seller at the ticket window. I always get the feeling that there are other good events that I can choose from, but I just do not know which ones are available. But the ticket seller usually is not willing or able to help much, especially when the ticket window is busy.*

A concise work activity note like this one may be enough:

> *I would like to be able to find my own events and not depend on the ticket seller to do the browsing and searching.*

7.2.4 Make Each Work Activity Note Complete

In your effort to be concise, do not sacrifice completeness. Retain the original meaning and remain true to the user's intentions.

- Fill in missing information.
- Include rationale where appropriate.
- Generalize a work activity note if you are sure it helps.

Include rationale where appropriate. If rationale information (why a user feels a certain way or does a certain thing) exists in the raw data, it should be considered to go along with the corresponding work activity note to make it more complete.

Example: Including Rationale in a Work Activity Note for the Ticket Kiosk System

An example of a note that includes rationale is:

> *I do not ask for a printed confirmation of my ticket transaction because I am afraid someone else might find it and use my credit card number.*

If there are two reasons in the rationale for the idea or concept in the note, make the notes elemental by splitting it into two notes. Repeat any context (e.g., rationale), as those two notes may eventually end up in different places.

Generalize cautiously. Because work activity notes are based on raw data, which can be very specific, be alert for the need to generalize to capture a more complete statement of the need being addressed. Be cautious about inventing new work activity notes that go beyond the spirit of the raw data.

Example: Generalizing a Work Activity Note for the Ticket Kiosk System

This same note in the previous example can safely lead you to compose a work activity note more generally about security of credit card information:

> *Must protect user credit card information.*

This can also lead to the following more specific work activity notes, while staying within the spirit of the original work activity note (although these may be better if considered as possible solutions):

> *Provide option of not printing confirmation of a transaction.*
> *Hide sensitive payment information (e.g., credit card number) in a transaction confirmation.*

7.2.5 Make Each Work Activity Note Modular

Modular in this context means that each note has enough information to stand on its own. It is a note that everyone can understand independently of all the others, allowing it to be rearranged, replaced, combined, or moved around independently of the others.

Retain context. If notes are split to be elemental, this means preserving any context that each split-off piece shares with its companions.

Example: Retaining Context When Splitting

This example shows you how *not* to make modular notes:

> *Important to potential customers: A method such as the one just above is essential for getting to the needed information the fastest way, especially if you already know the name of the event.*

Note how the reference to "A method such as the one just above" became unresolved after this note was split off from the preceding part.

Avoid indefinite pronouns. Make sure each work activity note does not contain unresolved pronoun references, such as he, she, they, them, this, or it, unless the referent is identified in the same note. State the work role that a person represents rather than using pronouns. Add words to disambiguate, and explain references to pronouns or other context dependencies.

Example: Avoiding an Unresolved Indefinite Pronoun in a Data Note

Suppose that, when another note was split to make it more modular, one of the resulting notes is:

> *He didn't ask me if I wanted to see a movie.*

At the time the original note was split, this part should be disambiguated to read:

> *The ticket seller did not ask me if I wanted to see a movie.*

Exercise 7.1 Compose Work Activity Notes for Your Product or System

Goal: Get practice in composing work activity notes from your UX research data (see Section 5.6.1.2).
Activities: If you are working alone, it is time for a pizza-and-beer UX research analysis party with your friends.

- Regardless of how you form your team, appoint a team leader and a person to act as note recorder.
- The team leader leads the group through raw data, composing work activity notes on the fly.
- Be sure to filter out all unnecessary verbiage, fluff, and noise.
- As the work activity notes are called out, the recorder types them into a document.
- Everyone on the team should work together to make sure that the individual work activity notes are disambiguated from context dependencies (usually by adding explanatory text in italics).

Deliverables: At least a few dozen work activity notes composed from your raw UX research data. Highlight a few of your most interesting work activity notes for sharing.
Schedule: Based on our experience with these activities, an hour should be enough time.

7.3 SYNTHESIZE UX RESEARCH DATA FOR INSIGHT AND UNDERSTANDING

Next, we need to make the UX research data useful, eventually able to influence the UX design. This entails synthesizing the data into models that we will describe in this section.

7.3.1 Synthesis Begins to Play a Role in UX Research

Synthesis is one of the most important tools in the UX technique toolbox (see Section 2.3). The general meaning of "to synthesize" is to combine things (e.g., information) you already have to create something new (e.g., an insight).

In UX research, the goals of synthesis are:

- To combine raw UX research data from multiple sources
- To identify unifying and underlying themes about the user work practice and work domain
- To achieve an integrated understanding of UX design needs

It is in the building of the WAAD (Section 7.4) and UX research data models (see Chapters 8 and 9), and the deduction of user stories and requirements (Chapter 10), that you are synthesizing—that is, creating and discovering artifacts, representations, and relationships, thus transforming raw UX research data into an understanding of UX design needs.

7.3.2 UX Research Data Present Multiple Overlapping Perspectives

The raw work activity data or distilled (elemental) work activity notes describe multiple and often overlapping perspectives of the same phenomena in the work practice. For example, a note about a Middleburg University Ticket Transaction Service (MUTTS) ticket seller finding the task of describing available entertainment options to a buyer during rush hour to be stressful offers insights into work role definition, task models, social models, breakdown or barriers during sequence models, and possibly an entry in the WAAD (Section 7.4). Sometimes, even if the content of a work activity note is absorbed through synthesis into a model (say, a hierarchical task inventory [HTI] model; see Section 9.2), it can also appear in a WAAD, where it can represent a different, or even a redundant, perspective of the work practice.

In this context, the overlapping of perspectives makes a good connection with synthesis, which is all about bringing together different perspectives to form a rich, integrated picture.

7.3.3 UX Research Data Are Incomplete

In data gathering (see Chapter 6), we use inquiry and observation to learn about this work practice and work domain, capturing what we learn through raw UX research notes and sketches. Because this data gathering does not cover all aspects of the work—only those that we were able to gather through the UX research process—what we capture is incomplete.

Because subsequent work activity notes are created from incomplete raw data, these do not describe or explain the totality of work, either. Completeness is not a goal (or an expectation) when we create work activity notes or a WAAD.

7.3.4 Modeling Supports Completeness

We create models of various aspects of this work in the synthesis chapters (Chapters 8 and 9), and there is an implied need for at least a degree of completeness. We are helped by the nature of modeling because it naturally forces completeness in thinking. Models show up missing information in any dimension of the problem space. By the time we finish synthesis, we should have a full understanding of the work (in theory). In fact, this is a litmus test for the success of UX research analysis and synthesis.

What is the source of the information that aids completeness of these models? The answer is in two places: (1) synthesis, extrapolation, and deduction by the researcher; and (2) the validation activities we do at each stage with users. This validation is aided by being in an agile context, where we have close collaboration with users or their proxies.

7.3.5 Combine Synthesis in Parallel With Work Activity Note Distillation

For clarity, we describe the following in separate linear steps in Sections 7.2 and 7.3:

- Creation and writing of work activity notes
- Utilization of work activity notes to synthesize models of the data

Experienced UX practitioners will find it expedient to perform these two activities in close combination. Immediately after creating a work activity note, or even as you are creating it, you can already begin to think about synthesizing its content as described in the rest of this chapter.

7.3.6 Synthesize Elemental Parts of Work Activity Notes

In Section 7.2.1, a goal was for work activity notes to be elemental, each about exactly one simple basic concept. Because of the nature of how most UX concepts are intertwined, however, we cannot expect to create work activity notes that are truly elemental. However, work activity notes themselves are sufficiently small and focused, so you can easily identify elemental parts to be synthesized.

Consider this simple work activity note:

[As a ticket buyer] I want to be able to browse and search for events by myself.

We put the "As a ticket buyer" in square brackets because it is unlikely a user would say that, but let's assume a UX practitioner added this user work role information. We can tease out these elemental parts:

- Functionality expressed in terms of two user tasks—browsing and searching
- The ticket buyer, the user work role associated with these tasks
- Events, the object of the browsing and searching tasks

7.3.7 Choose Synthesis Sites

Work activity note synthesis is about making choices. The models and the WAAD are potential choices of sites for synthesizing the information contained in a work activity note. The question involved in making these choices is whether the work activity note being synthesized substantially or materially contributes to the aspect of the work practice represented by a given model or location in the WAAD.

As you synthesize the concept embodied in each work activity note, the different data models chosen by the UX practitioner provide overlapping perspectives for interpreting and understanding the story told by that work activity note.

In general, a single work activity note that describes a user role, goal, or task can contribute to multiple models for user work roles, user classes, personas, task structure, task sequences, the flow model, and the WAAD. These are all choices for synthesis that the UX team must make, depending on how useful each model would be in contributing to the overall understanding of the work practice.

Synthesizing the elemental parts of the example work activity note of Section 7.3 leads us to these choices of where to look for synthesis opportunities:

- In the user work role model, check if ticket buyer exists; if not, then add it.
- Because the HTI, or hierarchical task inventory, model (Section 9.2.2) model is usually organized by user work role, see if the ticket buyer is a role within the HTI; if not, then add it.
- Check the HTI under the role of ticket buyer for the existence of browsing and searching tasks. If there, then see if any extra information in the work activity note may enhance the task representations and add/merge it into the HTI model.
- See if the task models (HTI and task sequence) or requirements include events as the object, and add/merge as needed.

This is a very simple example of how the data models are filled out and built into useful tools of understanding.

7.3.8 Be Selective in Choosing Models

One thing that makes choosing synthesis sites work without too much effort in practice is the project context. If the context is lightweight (e.g., working on a domain-simple, interaction-simple quadrant of the complexity matrix; see Section 3.2.1), then we may hold off on doing much of the detailed modeling of Chapters 8 and 9. We will likely do only work activity notes and the WAAD, structuring the higher-level branches of the WAAD along the lines of the types of data models, such as tasks, roles, and physical layout issues. So, while we describe the high rigor process here, in practice we rarely need to traverse the problem space so many times along so many dimensions.

On the other hand, if the project context is more at the opposite end of the complexity matrix, and the team does not have expertise in the domain, we may need to look at the work practice and work domain through the lenses of multiple models.

Even in this complex end of the matrix, we will almost never do all the models. In practice, we usually only model those areas that we do not understand well, so we will most likely rely on only a few models for most contexts.

7.4 ORGANIZE UX RESEARCH DATA

Aside from making data models, you can organize your data to promote easier understanding, techniques for which include:

- ***Hierarchical indented lists.*** For small amounts of data and simple categories, hierarchical lists (see Section 2.3.2.1) are an economical approach to organization. The highest level category is represented at the top of the hierarchy. Next-level sub-categories are indented under their corresponding parent categories.
- ***Card sorting.*** Card sorting (see Section 2.3.2.2) is a technique for ordering cards annotated with concepts into categories based on shared characteristics. Card sorting is usually limited to small amounts of data (e.g., to organize menu structures in desktop applications). Designers list all the actions the system should support, then they print each action on a card. Users or other stakeholders are then asked to organize them into appropriately related groups.
- ***Concept/mind mapping.*** Mind mapping[2] and concept mapping[3] (see Section 2.3.2.3) are graphical techniques for drawing labeled images centered on concepts and relationships among them as a way to explore new ideas or concepts relative to other known ideas and concepts.
- ***Affinity diagrams.*** Affinity diagrams (see Section 2.3.2.4) are used to organize large amounts of data, grouping data hierarchically by relatedness (affinity) and common attributes.

The first three techniques require no further explanation here. In the next section, we walk you through an approach to construct a WAAD.

7.5 CONSTRUCT A WORK ACTIVITY AFFINITY DIAGRAM TO ORGANIZE UX RESEARCH

We organize the work activity notes of our UX research data, especially notes about qualitative and open-ended feedback, into our variation of an affinity diagram (i.e., a WAAD). A WAAD identifies key (abstract) themes in the work practice and relationships among them. As an example, some of these themes may be about the various types of users who operate in the work practice we are studying. These insights overlap with the more concrete and complete perspectives on work roles we will see in the data models, such as user work roles (see Section 8.2), personas (see Section 8.3), and concerns about social aspects of the work practice (see Section 9.4).

The WAAD can afford visualization of the user's work context and help generalize from instances of individual user work activities to broader representations of a work practice or domain. WAADs serve the data organization needs over a broad range of project size, complexity, and need for rigor.

[2]https://en.wikipedia.org/wiki/Mind_map
[3]https://en.wikipedia.org/wiki/Concept_map

7.5.1 Affinity Diagrams

An affinity diagram is a bottom-up hierarchical technique for organizing and grouping large amounts of disparate qualitative data to highlight the issues and insights in them in a visual display. The technique came from Jiro Kawakita, a Japanese anthropologist, as a means to bring to focus large amounts of data from the field (Kawakita, 1982). Data notes about a similar idea are said to have an affinity for each other and are therefore grouped together by affinity. We adapt affinity diagramming as a technique for building a WAAD to organize and group the issues and insights by similarities and themes to highlight common work patterns and design needs.

WAAD building in most real projects tends to be agile and very informal. Start by printing out your work activity notes in a card-size format or use handwritten work activity notes.

7.5.2 Compartmentalize the WAAD, Separating It by User Work Roles

If the complexity of our project requires it, we can build a WAAD to organize the work activity notes coming from the UX research data. In many cases, there are not enough work activity notes to justify a complicated formal process of WAAD building; you are encouraged to use the work activity notes directly in building data models and writing requirements.

If the complexity warrants, we begin at the highest level by dividing the WAAD into multiple separate WAADs, one for each user work role (Fig. 7.3). This separation allows analysis and design for one work role at a time.

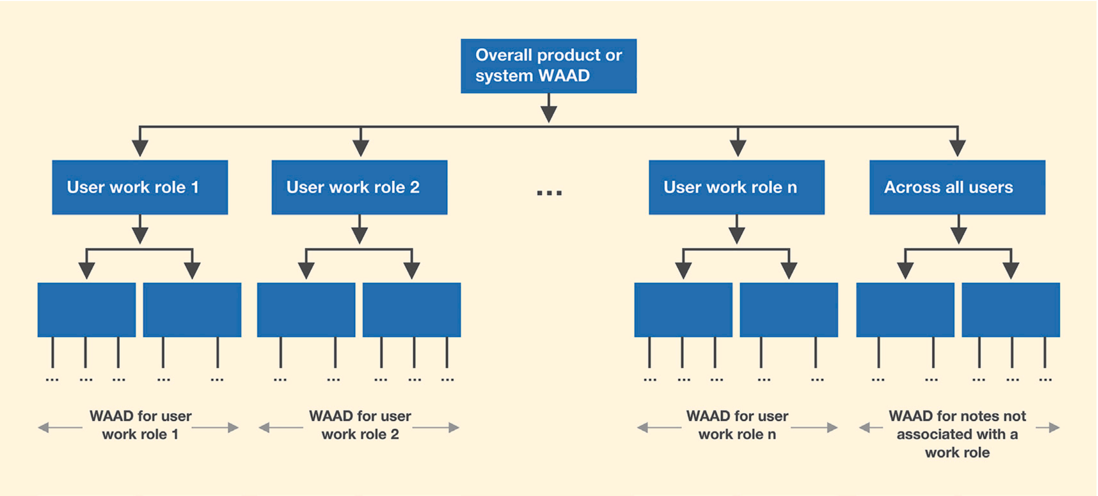

Fig. 7.3
Work activity affinity diagram (WAAD) *compartmentalized by user work roles.*

7.5.3 Growing the WAAD—A Bottom-Up Process

If two work activity notes are sorted to the same node (group), we say they share the same affinity. All the work activity notes attached to the same node (i.e., under the same label) are considered to be a group. Consider the following when building your WAAD:

- Consider each input work activity note in turn.
- Decide which user work role is the best fit, or start a new work role group.
- If the input work activity note fits an existing topic under this work role, then add it to that group (i.e., under the node with this group label). This is how the WAAD grows in depth.
  - Adjust the group label as necessary to encompass the new note.
- If the note to be posted does not fit the theme of any existing group, then:
  - Start a new group under the relevant role with this work activity note (this is how the WAAD grows in breadth)
  - Add a topic label to identify the theme of this new group (see next section on labeling)
- Continue in this fashion, growing and evolving groups of work activity notes.

7.5.4 Labels for Groups of Notes

Each group label must describe the affinity that ties the entire group together.

- A precise group label is important for capturing the essence, gestalt, or meaning of the group.
- A highly descriptive group label replaces your notes because it provides complete information about the group.

As an example of the importance of precision in labels, a team in one of our sessions used the label, "How we validate information," when they really needed the more precise label, "How we validate input data forms." The difference between the two is subtle, but it is important to pin down the intended affinity for that group.

Example: Growing Labels With Growing Groups

The purple label in Fig. 7.4 shows terms that were added in the course of evolution within a group of work activity notes for the Ticket Kiosk System.

The group label started out as "Security concerns." Notes about privacy and trust were added to this group because those concepts have an affinity with security concerns.

Fig. 7.4

A topical label that has grown in scope during affinity diagram building.

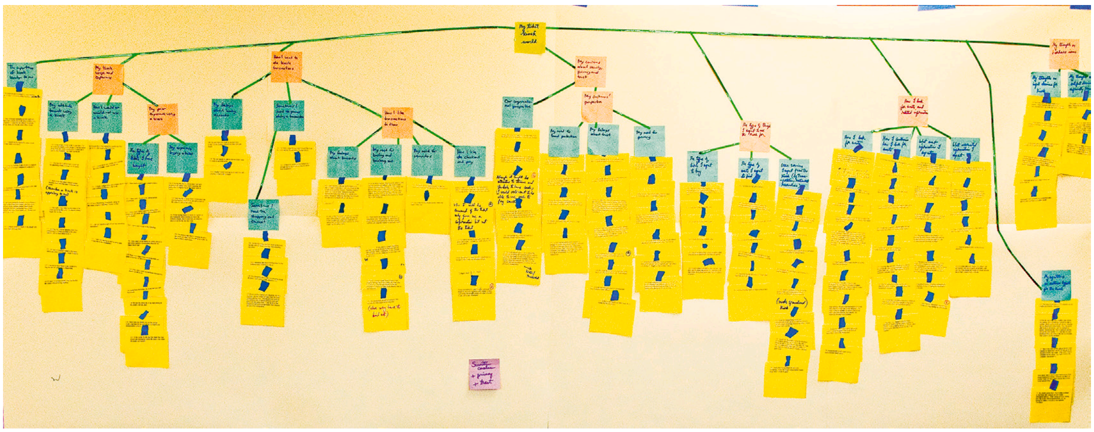

Fig. 7.5

The WAAD that we built for the Middleburg University Ticket Transaction Service example.

7.5.5 Splitting Large Groups

As groups can get too large and the topic too broad to be useful, reduce scope by splitting groups.

7.5.6 Continue Organizing Groups Into a Hierarchy

Keep refining and reorganizing the groups and labels. The order and organization of the groups and their labels should be considered flexible and malleable.

Fig. 7.5 shows a large part of the overall WAAD we built for MUTTS.

Example: Organizing Ticket Kiosk System Work Activity Notes

Returning to the running example of the Ticket Kiosk System, we provide examples of work activity notes selected from several hundred real UX research work activity notes for the sample system, along with categories that emerged from data in the process. We show how the notes can be organized into the hierarchical categories of a WAAD. This is also an example of a textual representation of a WAAD.

The contents in this example are real work activity notes obtained by multiple UX researchers. Any flaws you spot in these notes make them like notes you might see in a real project, which is what they are. The work activity note itself is in italics to set it off from the structural elements.

- *The kiosk should remind people to take the ticket and credit card (at the end of the transaction).*
- *Need graphical display of seating arrangements (text description of types of seating not meaningful enough to decide).*
- *Sort by categories (sports, movies, etc.), date/time, price, location. User drills down into hierarchical structure.*
- *It will be very important to have a confirmation page showing exactly what I'm buying before I pay (as good online sites do).*
- *I would like to be able to browse by different topics (type of event). For example, in big cities, even in music there will be different genres.*
- *I would use the kiosk if it is affiliated with Middleburg University and MU sends an email informing us about it.*
- *Idea: Although marketing people might not think to put a kiosk next to the ticket window at the ticket office, it would be very helpful for people who get there and find the ticket office closed.*
- *High-priority need/want: Calendar on front page. Pick date and click "What's going on today?" or "What's going on for this date?"*
- *It's complicated to recognize bills (cash) and dispense change.*
- *A shopping cart model will facilitate buying different types of tickets in one go.*
- *Having cash inside the kiosk would be attractive to thieves and vandals.*
- *A kiosk has to be presented in a professional way; it needs to look official.*
- *It (the kiosk) should not look cheap like a holder for free brochures.*
- *Also, need video trailers. Downside: expensive to produce and take too long, holding up other people in line.*
- *The kiosk should have a 24/7 customer service number prominently displayed.*
- *Kiosk needs a concept of a shopping cart, so a customer can buy tickets for more than one event without starting over. This should even work for two different sets of tickets (different seats and prices) for the same event.*
- *I want to be able to look at seating options versus prices.*
- *Choosing date and time. Kiosk gives options for current day, this coming weekend, within the current week, etc. Future dates selected from an interactive calendar image.*
- *Regardless of the difficulties it might present to designers, I would still want to be able to use cash to pay sometimes.*

- *To use at Metro station: Need a quick cancel to bail you out immediately and go to the home screen without leaving a trace of what you were doing (for privacy with respect to the next customer).*
- *To recover later, need a quick path to a specific item. Maybe use event ID#s to get there directly next time, like using a catalog or item number in search for online shopping.*
- *I'd want to be able to use cash, credit card, or debit card to pay. I'd insert or tap my credit card as I do at a gas station.*
- *I want a ticket option allowing several friends to sit together.*

Here are the previous work activity notes set in a very simple WAAD structure:

- Interaction design
  - Transaction flow
    - *The kiosk should remind people to take the ticket and credit card (at the end of the transaction).*
    - *It will be very important to have a confirmation page showing exactly what I'm buying before I pay (as good online sites do).*
    - *It is essential to provide a way to bail out of a transaction.*
    - *If the user is at Metro station and a train is coming, need a quick cancel to bail out immediately and go to the home screen without leaving a trace of what you were doing (for privacy with respect to the next customer).*
    - *To recover later, need a quick path to a specific item. Maybe use event ID#s to get there directly next time, like using a catalog or item number in search for online shopping.*
- Venue representation
  - Visual display of seating arrangements
    - *Need graphical display of seating arrangements (text description of types of seating not meaningful enough to decide)*
    - *I want to be able to look at seating options versus prices.*
    - *I want a ticket option allowing several friends to sit together.*
- Attributes of events
  - Video trailers
    - *In some cases I would like to see video trailers before making my choices.*
    - *Downsides: Expensive to produce. Trailers take too long to view, holding up other people in line.*
- Special features
  - Shopping cart concept
    - *Kiosk needs a concept of a shopping cart, so a customer can buy tickets for more than one event without starting over. This should even work for two different sets of tickets (different seats and prices) for the same event.*

        - *A shopping cart model will facilitate buying different types of tickets in one go.*
    - Help, customer support.
        - *The kiosk should have a 24/7 customer service number (and kiosk ID number) prominently displayed.*
- Payment methods
    - Credit card usage
        - *I'd want to be able to use cash, credit card, or debit card to pay. I'd insert or tap my credit card as I do at a gas station.*
    - Cash transactions
        - *It's complicated to recognize bills (cash) and dispense change.*
        - *Regardless of the difficulties it might present to designers, I would still want to be able to use cash to pay sometimes.*
- Searching and browsing
    - Calendar-driven choices
        - *High-priority need/want: Calendar on front page. Pick date and click "What's going on today?" or "What's going on for this date?"*
        - *Choosing date and time. Kiosk gives options for current day, this coming weekend, within the current week, etc. Future dates selected from an interactive calendar image.*
    - Event-driven choices
        - *I would like to be able to browse by different topics (type of event). For example, in big cities, even in music there will be different genres.*
- Branding
    - Kiosk appearance
        - *A kiosk has to be presented in a professional way; it needs to look official.*
        - *It (the kiosk) should not look cheap like a holder for free brochures.*
- Information architecture
    - Structure, organization by category
        - *Sort by categories (sports, movies, etc.), date/time, price, location. User drills down into hierarchical structure.*
- User ticket-shopping behavior
    - Likelihood of using a kiosk
        - *I would use the kiosk if it is affiliated with Middleburg University and MU sends an email informing us about it.*
- Business issues
    - Kiosk locations
        - *Idea: Although marketing people might not think to put a kiosk next to the ticket window at the ticket office, it would be very helpful for people who get there and find the ticket office closed.*
        - *Having cash inside the kiosk would be attractive to thieves and vandals.*

7.5.7 Use Technology to Support WAAD Building

A very large wall space works well for teams doing WAAD building (Fig. 7.6) with slips of paper.

For teams that do WAAD building frequently, there are higher-tech WAAD-building tools. Fig. 7.7 shows a team at Virginia Tech using affinity diagramming software on a high-resolution large-screen display as an alternative to paper-based work activity note shuffling (Judge et al., 2008). Work activity notes residing on a laptop can be moved around by touching and dragging.

Fig. 7.6

Team working to build a WAAD.

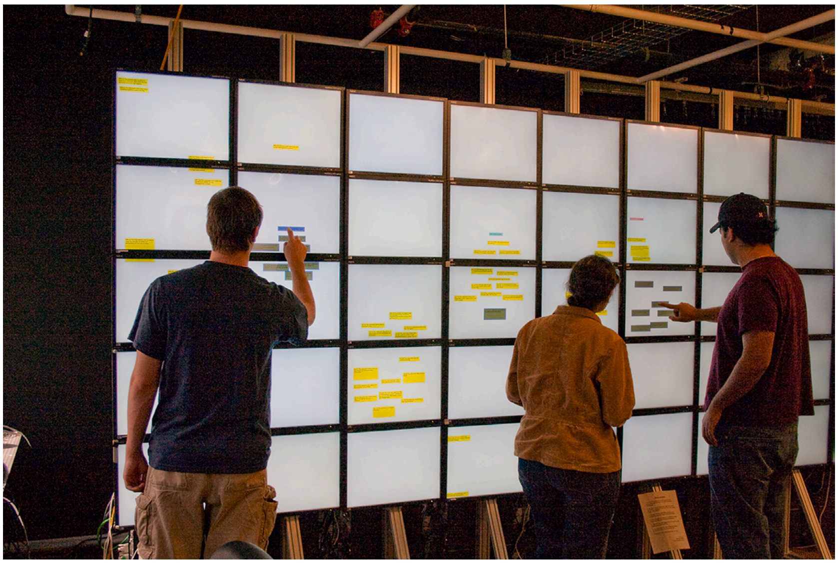

Fig. 7.7

Building a WAAD on a large touchscreen.

It would be nice to have a WAAD-building app to work in coordination with the large touchscreen to manage WAAD content in a computer-readable representation.

7.5.8 Lead a Walkthrough of the WAAD to Get Feedback

If feasible in your budget and schedule, bring in users, clients, and other stakeholders to walk them through your WAAD to share your findings and get feedback.

Exercise 7.2 WAAD Building for Your Product or System

Goal: Get practice in building a WAAD to organize your work activity notes by category.

Activities: If you are working alone, you may have to have friends over for pizza, again.

Using the work activity notes created in the previous exercise, do your best to follow the procedure we have described in this chapter for WAAD building.

Take digital photographs of your work process and products, including the full WAAD, some medium-level details, and some closeups of interesting parts.

Hang them on your fridge with magnets.

Deliverables: As much of the full WAAD for your system as you were able to produce, plus the digital photos you took of it.

Schedule: This is one of the more time-consuming exercises; expect it to take a few hours.

7.6 SEGUE

In this chapter, we discussed how to analyze UX research data. In the next chapter, we discuss traversing the work practice space through synthesized models, each providing a different perspective into this space.